Week #8: [Osteology & Joints of the Distal Upper Extremity]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

- Introduction to Module

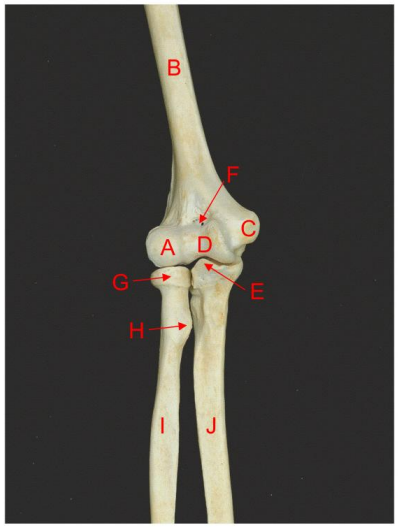
Osteology & Joints of the Distal Upper extremity

- Anatomy of Bones & Joints (41 min)
- Approach to Elbow Joint Problems (27 min)

Anatomy of Bones & Joints

Review: Elbow Osteology

Trochlea
Capitulum
Medial Epicondyle
Humerus
Radial Head
Coronoid
Coronoid Fossa
Ulna
Radius
Radial Tuberosity



I. Osteology:

A. Humerus – only bone of the arm

1. **Radial (spiral) groove** – located on posterior shaft; ***radial nerve*** and ***profunda brachii artery*** run along groove
2. **Medial epicondyle** – medial protuberance at the distal end of the humerus; origin of the ***common flexor tendon*** for the forearm flexors; ***ulnar nerve*** courses posterior to it
3. **Lateral epicondyle** – lateral protuberance at the distal end of the humerus; origin of the ***common extensor tendon*** for the forearm extensors
4. **Capitulum** – lateral condyle; articulates with the head of the radius
5. **Trochlea** – medial condyle; articulates with the *trochlear notch* of the ulna
6. **Coronoid fossa** – depression superior to the trochlea anteriorly; articulates with the coronoid process during full flexion
7. **Olecranon fossa** – depression superior to the trochlea posteriorly; articulates with the olecranon process during full extension

Clinical Note: Fractures of the **surgical neck** of the humerus, often caused by low energy falls to the hand are especially common in elderly individuals with osteoporosis. This type of fracture may result in damage to the ***axillary nerve.*** Transverse fractures of the shaft of the humerus frequently result from a direct blow to the arm. This may result in injury to the ***radial nerve*** and/or ***deep brachial artery.*** Fractures of the distal humerus may damage the ***median nerve.*** Fractures of the **medial epicondyle** may result in damage to the ulnar nerve.

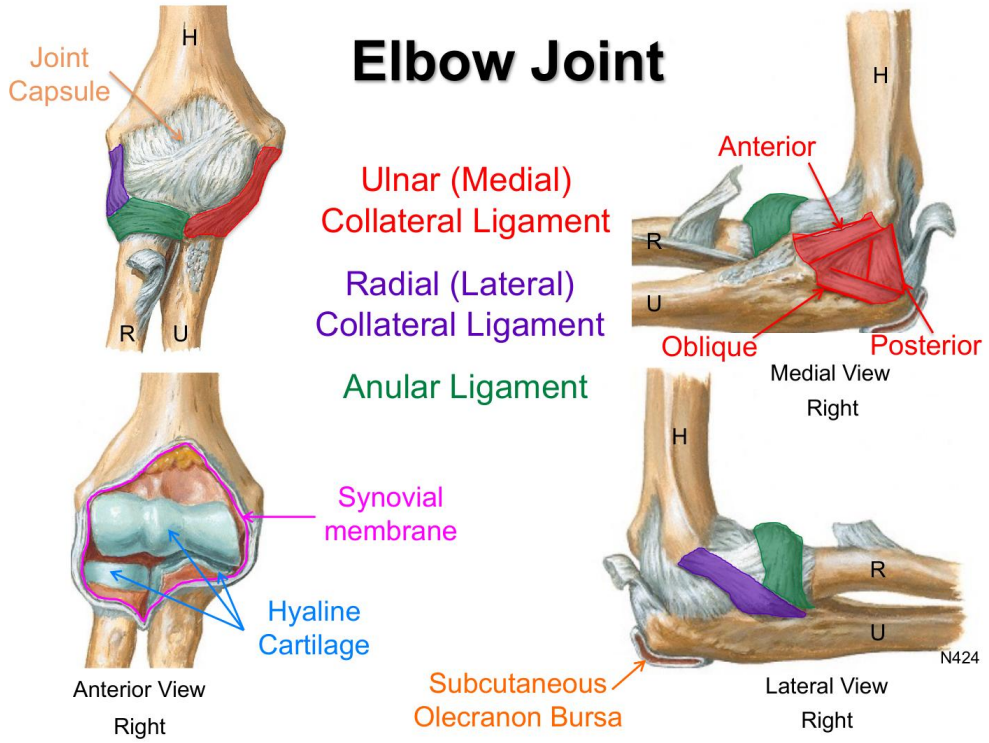
B. Radius – lateral and shorter bone of the forearm

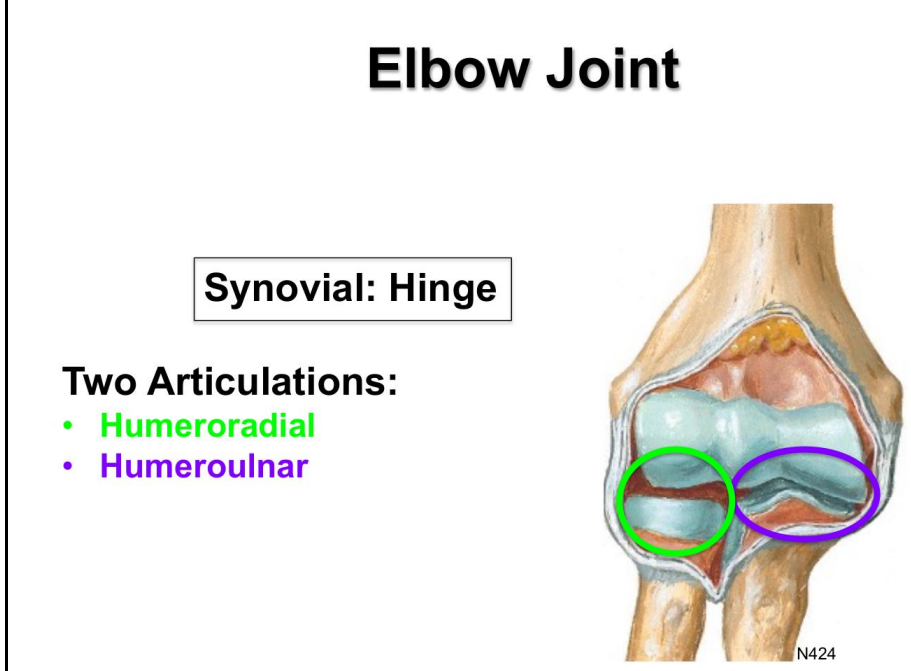
1. **Head** – proximal portion that articulates with the capitulum of the humerus and radial notch of the ulna; pivots within the ***annular ligament***
2. **Neck** – narrow portion between the head and the radial tuberosity
3. **Radial tuberosity**- medial projection; insertion site for ***biceps brachii***

C. Ulna – medial and longer stabilizing bone of the forearm.

1. **Olecranon process** – bony eminence at posterior aspect of proximal ulna; insertion site for ***triceps brachii***
2. **Coronoid process** – triangular protuberance at anterior aspect of proximal ulna; insertion site of ***brachialis***
3. **Trochlear notch** – large depression between olecranon and coronoid process; articulates with trochlea

4. **Radial notch** – concavity on lateral side of coronoid process that articulates with the head of the radius





II. Joints of the Forearm

A. Elbow Joint

1. ***Type:*** hinge type of synovial joint; uniaxial
2. ***Articulation:*** humeroulnar (trochlea of humerus; trochlear notch of ulna) and humeroradial (capitulum of humerus; head of radius)

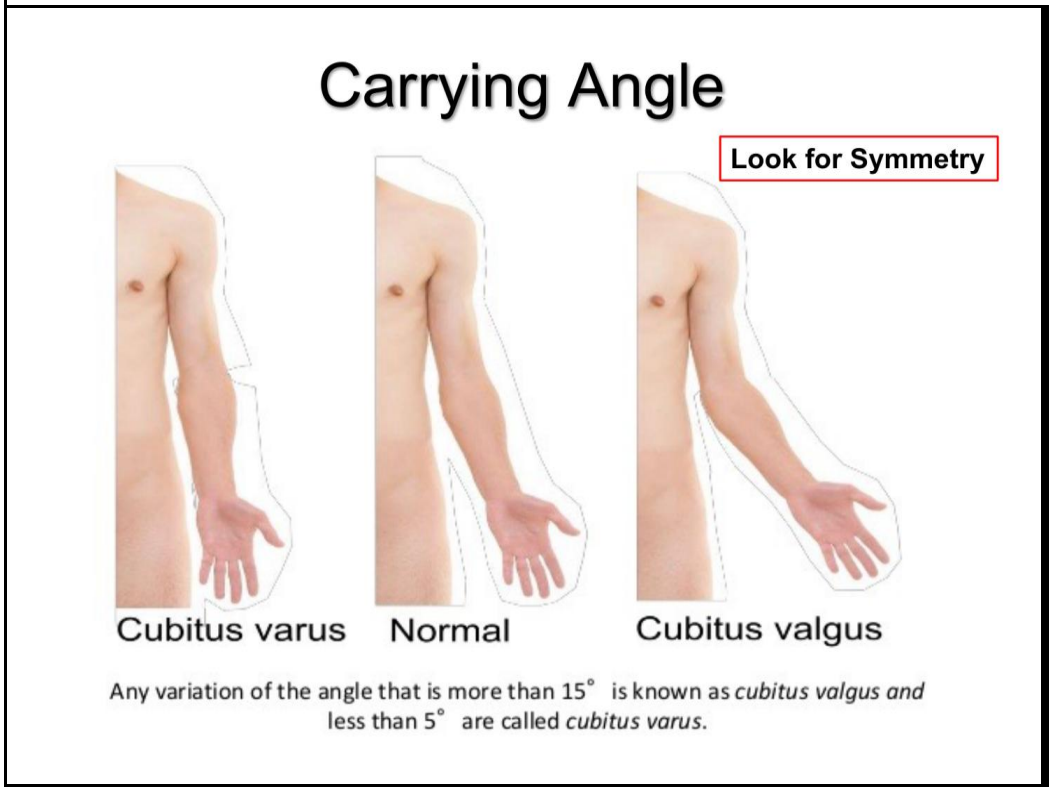
Note: The elbow joint capsule is weak anteriorly and posteriorly but reinforced medially and laterally by collateral ligaments; muscles help to strengthen the joint anteriorly and posteriorly; the elbow has several prominent bursae (i.e. subcutaneous olecranon bursae, biceps bursa) that help to facilitate movement.

3. ***Ligaments:***
   - ***a.*** Radial collateral ligament: extends from lateral epicondyle of humerus to the head of the radius where fibres blend with annular ligament of radius
   - ***b.*** Ulnar collateral ligament: extends from medial epicondyle of humerus to coronoid process and olecranon of ulna. Consists of three bands: anterior (cord-like, strongest), posterior (fan-like, weakest), and oblique (deepens socket for trochlea of humerus)

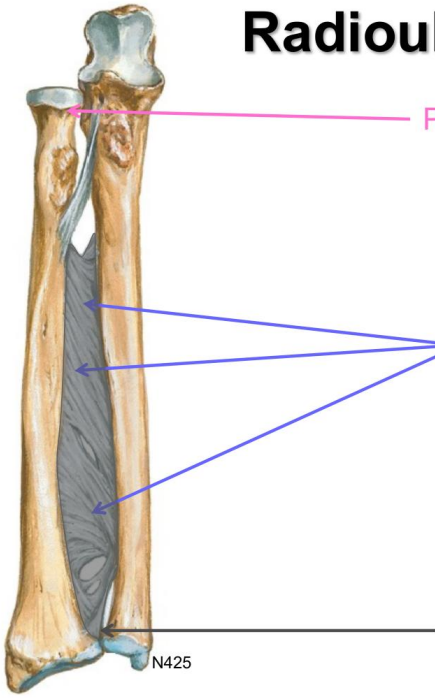
Clinical Note: replacement of a damaged ulnar collateral ligament with a muscle tendon (Tommy John surgery) somewhat common in baseball pitchers; 1-2 years of physical therapy required

4. ***Movements:*** flexion/extension; review primary muscles involved.
5. ***Blood supply:*** many arteries participate in supplying the elbow joint, including the superior and inferior ulnar collateral a., the anterior and posterior ulnar recurrent a., the radial collateral br. of the deep artery of the arm, and the recurrent radial a.

Clinical Note: posterior dislocation of the elbow joint most common; may be associated with torn joint capsule, ruptured collateral ligaments and bone damage. Ulnar nerve is at risk due to its position posterior to the medial epicondyle.

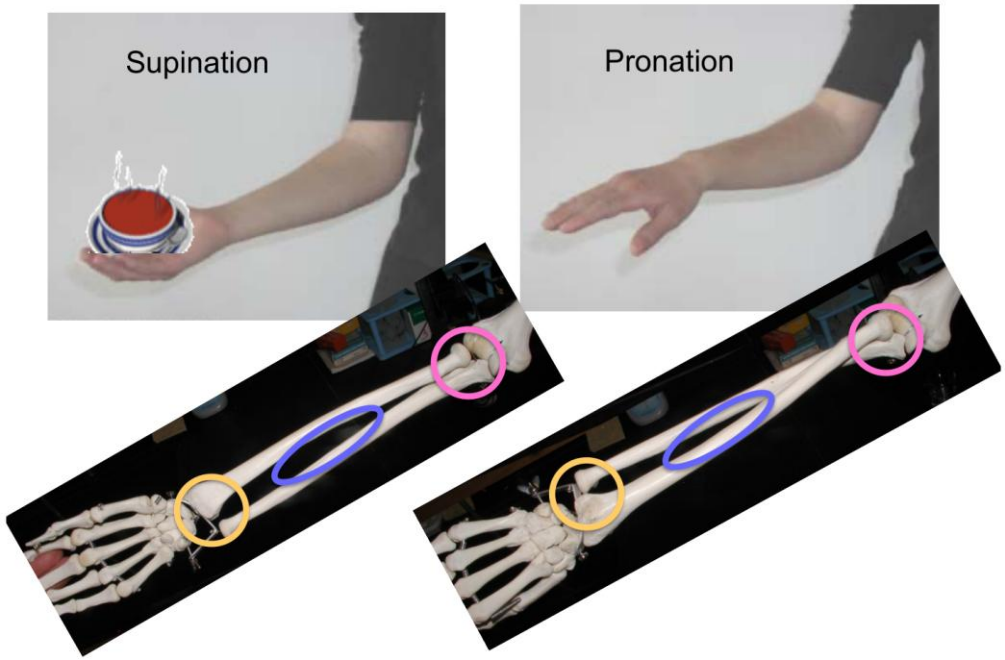


Radioulnar Joints

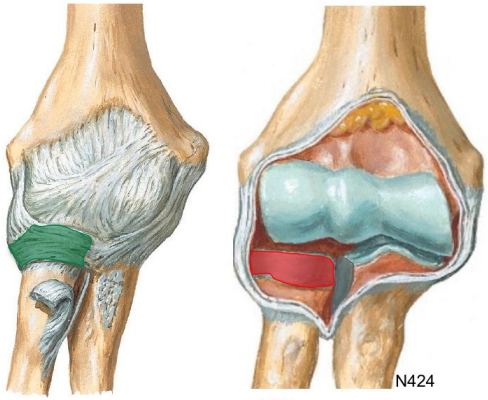


- Connected via the interosseous membrane

Movement of Radioulnar Joints

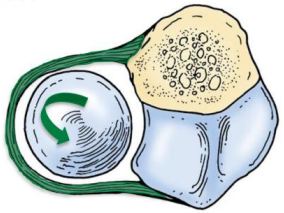


Proximal Radioulnar Joint



Synovial: Pivot Joint

- Articulation of head of radius & radial notch of ulna
- Located within loose fibrous capsule/synovial membrane of elbow joint



- Anular Ligament allows rotation

B. Proximal Radio-Ulnar Joint

1. ***Type:*** **pivot** type of synovial joint; uniaxial
2. ***Articulation:*** **head of the radius; radial notch of the ulna;** located within fibrous joint capsule of the elbow joint; shares a synovial membrane with the elbow joint
3. ***Ligament:*** the ring-like annular ligament of the radius encircles head of radius
4. ***Movements:*** supination/pronation; head of the radius rotates within annular ligament

Clinical Note: **"Nursemaid's Elbow"** (subluxation of the proximal radio-ulnar joint); Common in preschool aged children, due to a pulling of the distal aspect of the child's upper limb; typically painful due to pinching of the well-innervated annular ligament and/or joint capsule.

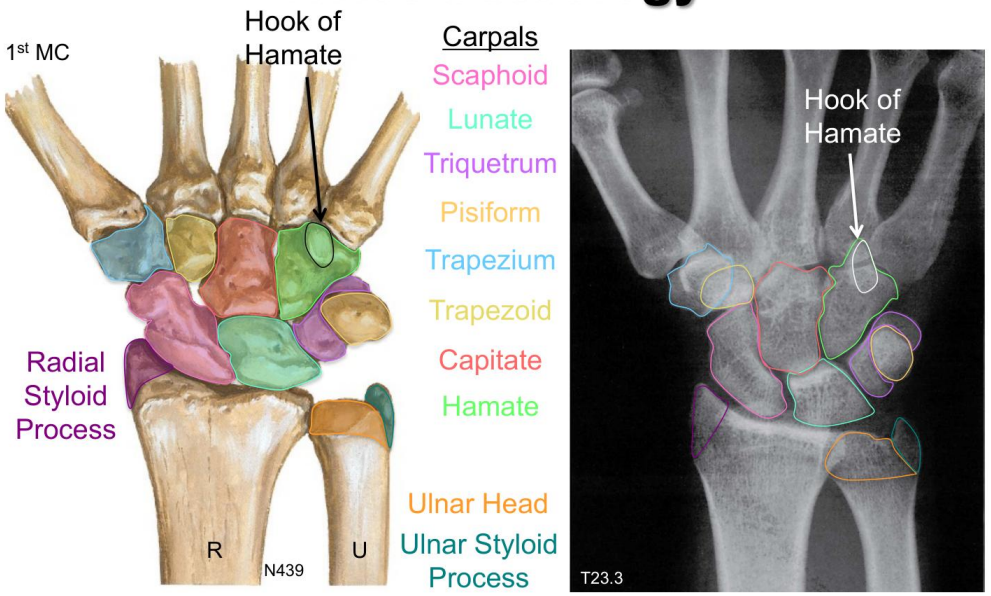
III. Wrist Osteology:

A. Carpals

1. Scaphoid – most commonly fractured carpal bone
2. Lunate – most commonly dislocated carpal, although uncommon if it does occur it may compress median nerve within the carpal tunnel
3. Triquetrum
4. Pisiform – forms ulnar canal with the hook of the hamate

5. Trapezium
6. Trapezoid
7. Capitate
8. Hamate – forms ulnar canal with pisiform
B. Hand
1. Metacarpals – 5 bones; each with a base, shaft, and head
2. Phalanges – thumb – 2; other digits – 3; each with a base, shaft, and head
a. Proximal phalanx – largest
b. Middle phalanx – intermediate in size
c. Distal phalanx – smallest
1. The phalanges serve as the insertion sit for many forearm muscles

Wrist Osteology

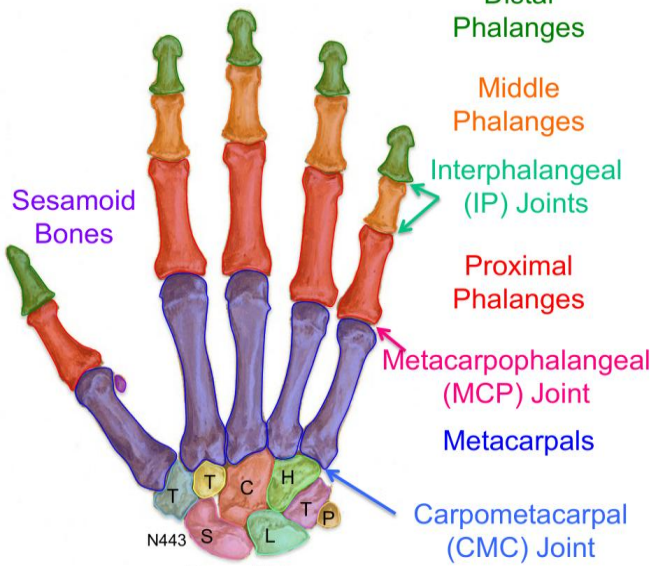


Carpals
Scaphoid
Lunate
Triquetrum
Pisiform
Trapezium
Trapezoid
Capitate
Hamate

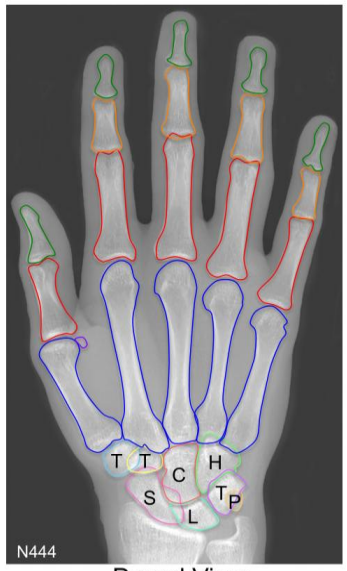
Ulnar Head
Ulnar Styloid Process

Some Lovers Try Positions That They Can't Handle

Hand Osteology



Dorsal View

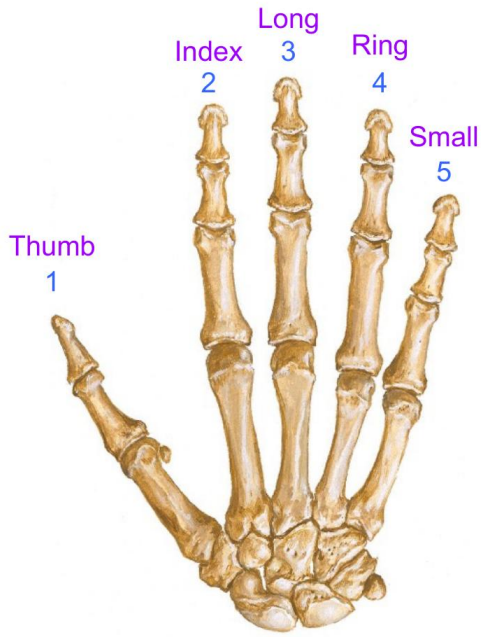


Dorsal View

Digits

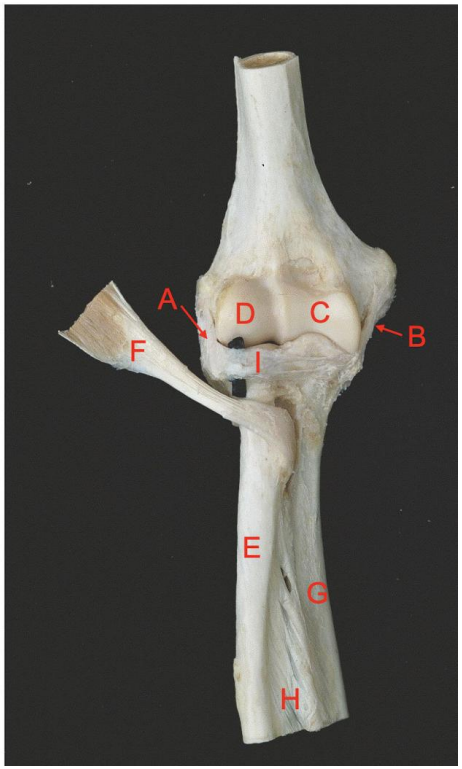
Clinical

Anatomical



Knowledge Check

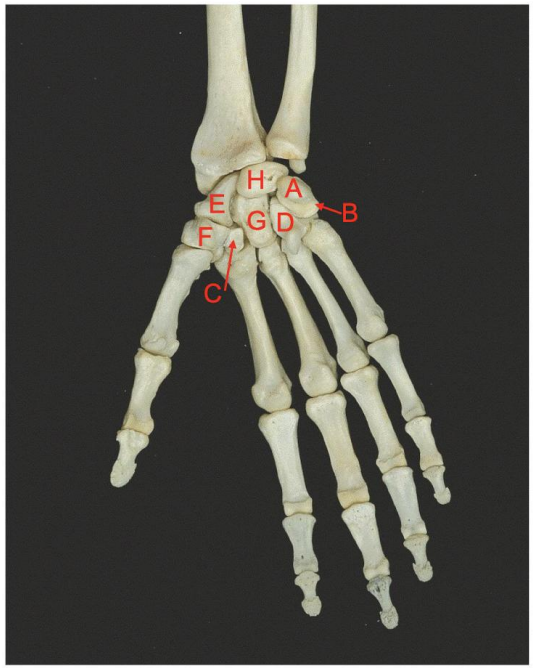
Bicipital Tendon (F)
Ulnar Collateral Ligament (B)
Capitulum (D)
Ulna (G)
Trochlea (C)
Interosseous Membrane (H)
Radial Collateral Ligament (A)
Radius (E)
Anular Ligament (I)



Knowledge Check

Trapezium (F)
Triquetrum (B)
Hamate (D)
Capitate (G)
Trapezoid (C)
Lunate (H)
Pisiform (A)
Scaphoid (E)

Anterior View of Right Hand



Osteology & Joints of the Distal Upper Extremity

Approach to Elbow Joint Problems

Clinical Approach

History	Physical	Management Approach
• Look for patterns!! • OLDCARS • **Acute** (traumatic, inflammatory) vs **chronic** (degenerative, overuse) • Overlay your knowledge of anatomy (which structures are medial, which are lateral, etc)	• LOOK at it • Is it swollen? Are there obvious deformities? • FEEL it • Where does it hurt to touch? Does it feel hot? • MOVE it • Is ROM restricted? • Specific testing maneuvers for specific structures • Don't forget the nerves!	• What (if any) investigations will help you? • If no red flags, apply your pattern • Educate • Non-pharmacologic • Pharmacologic • Interventional • Remember your patient is a person!

Differential Diagnosis for Elbow Problems

By Anatomic Structure	By Location
• Bone/ligament/joint • Arthritis • Degen, post-traumatic, inflammatory • Subluxation/dislocation • Sprain/tear • UCL • Fracture • Muscle/tendon • Tendinopathies • Medial epicondylosis (golfer's elbow) • Lateral epicondylosis (tennis elbow) • Distal biceps • Distal triceps • Other structures • Olecranon bursitis • Ulnar neuritis/neuropathy	• Anterior • Arthritis • Degenerative, post-traumatic • Inflammatory • Distal bicep tendinopathy • Medial • Medial epicondylosis • Ulnar neuritis/neuropathy • UCL sprain/tear • Lateral • Lateral epicondylosis • Posterior • Olecranon bursitis • Arthritis • Triceps tendinopathy (fracture could be anywhere)

Joint/Ligament Problems

- Issues with these anatomic structures may impact
  - Elbow ROM
  - Elbow stability
  - Elbow pain

Arthritis of the Elbow

- More details to come re: arthritis in general
- Broad categories
  - Degenerative (osteoarthritis)
  - Inflammatory (rheumatoid arthritis, gout, psoriatic arthritis)
    - Stay tuned for more info in rheumatology modules!
  - Post-traumatic
    - Essentially leads to earlier onset of OA, potentially in a different pattern

Osteoarthritis of Elbow

- Clinical Presentation
  - Degenerative history
    - Gradual onset
    - Somatic pain (achy/sore with acute exacerbations)
  - Mechanical symptoms
    - "Stiffness"
    - Loss of ROM
    - Locking
  - May see associated bony changes
  - Typical X-ray features (discussed in detail in rheum modules
    - Loss of joint space
    - Osteophytes
    - Subchondral sclerosis/cysts

Kaback, L.A., Green, A., & Blaine, T. CT Elbow Arthritis and total elbow Replacement.



Osteoarthritis of the Elbow

- Management (from Approach to Management of ALL THINGS MSK)
  - Education
    - Tell them what it is!
    - What's expected? Grumbling, long term course (no "cure" for OA)
  - Non-pharm
    - Modalities (ice)
    - PT to maintain ROM, OT to maintain function
  - Pharm
    - Topical anti-inflammatories
    - Oral acetaminophen, NSAIDs (consider side effect profile!), steroid injection
  - Intervention/surgery if intolerable end stage
    - Debridement
    - Arthroplasty (joint replacement)

Joint Subluxation/Dislocation

- Elbow has two significant joints to consider…
  - Elbow itself (humero-ulnar joint)
  - Proximal radioulnar joint
- Both can become dislocated, but present very differently!

Elbow dislocations

- Fall onto outstretched hand (FOOSH) may be most common mechanism
- Posterior dislocations most common – these aren't great!
  - Capsule rupture, ligament ruptures
  - Skin integrity
  - Nerve implications – ulnar nerve particularly at risk given location
  - Compartment syndrome
- Management – this is traumatic, this is a red flag condition!
  - Closed reduction and splinting at 90 deg
    - May not be possible if soft tissue is interposed
    - May not be able to maintain
  - Open reduction internal fixation (ORIF)

Radial head subluxation/dislocation

(ie. at proximal radioulnar joint…aka nursemaid's elbow)

- More common in girls and in the left arm
- Essentially exclusive to pediatrics (rare after age 5)
  - Annular ligament thinner and weaker distally in children, allows radial head to dislocate
- Presentation
  - Acute pain after longitudinal traction force through arm
  - Refusal to move/use arm; holds arm flexed against body
- Management
  - Imaging is only required to rule out other pathologies (ie. fracture)
  - Closed reduction (pronation and extension movement); no cast/orthosis required



Ulnar Collateral Ligament Injury

- UCL role? Stabilize medial elbow
- Injuries typically related to chronic/repetitive overuse – baseball pitcher
- Clinical presentation
  - Sudden tear: may have pop over medial elbow
  - Pain on palpation of medial elbow
  - Lack of end point with valgus stress
    - Excessive motion due to lack of ligamentous stability
  - Probably requires physical exam by sports med/ortho
- MRI may be helpful





Ulnar Collateral Ligament Injury

- Management (from Approach to Management of ALL THINGS MSK)
  - Education
    - Tell them what it is!
    - What's expected? Ligaments heal slooooooowly (if at all, depending on degree of tear)
  - Non-pharm
    - Modalities (ice)
    - PT – strengthen dynamic elbow stabilizers, proprioceptive activities
    - Orthoses – confer extra stability
  - Pharm
    - Topical anti-inflammatories
    - Oral acetaminophen, NSAIDs (consider side effect profile!)
  - Intervention/surgery
    - Reconstruction of ligament (Tommy John surgery) – only if necessary for functional goals

Tommy John





Photo credit: SportTechie

- Ulnar collateral ligament replacement surgery
  - Muscle tendon
    - Palmaris longus
    - Semitendinosus
    - Others…
- Treatment: years of PT

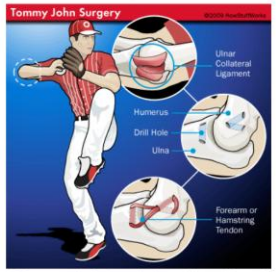


Photo credit: Orthopedic Specialists of Seattle

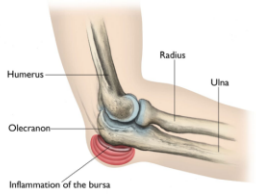
Olecranon Bursitis

- Not a bone/joint/ligament, but lives in close proximity to the joint
- Let's talk about bursitis for a sec…

Bursitis in General

- More than 150 bursae (thin, fluid-filled sacs) in the body
- Bursae are located between bone and soft tissue (including skin) and act as cushions while permitting tissues to move freely over underlying bone
  - Consider mobility of skin over elbow
- Normally, a bursa is flat, but accumulates fluid if irritated or inflamed, with potential symptoms:
  - Pain, focal tenderness
  - Pain, restricted ROM
  - Swelling if/when superficial

Olecranon Bursitis



- Most common superficial bursitis
- Etiology:

  a) Single hard trauma – fall on elbow

  b) Sustained / repetitive trauma – regular leaning on elbow
    - Develops over months
    - Consider at-risk populations or positions (elbow leaners)
      - Construction, plumbers, HVAC technicians in confined spaces, office jobs; some athletes, some sedentary positions

  c) Infection
    - Identifiable skin break (bite, scrape, etc) or non-identifiable skin injury
    - Bacterial infection causes usual bursitis symptoms plus erythema, systemic responses

How Can you Distinguish Them?

- Need to be able to identify the bad/red flag bursitis cases…
  - History will be helpful!
  - What about physical? They do tend to look different!

“Not Scary” Bursitis

- Often no investigations required
  - May consider X-ray if history of trauma
  - If it doesn’t behave normally, can consider US or MRI



Photo credit: By Harrygouvas, Attribution, https://commons.wikimedia.org/w/index.php?curid=9713

“Bad” Bursitis

- This is an infected bursitis!
  - Aspirate and choose antibiotics appropriately



Olecranon Bursitis

- Management (from Approach to Management of ALL THINGS MSK)
  - The following applies if you’ve ruled out red flags
    - Septic bursitis requires antibiotics +/- I&D depending on severity
    - Gout or inflammatory arthropathies (that make the whole elbow look swollen and might be confused for bursitis) require specific treatment
  - Education
    - Tell them what it is!
    - What’s expected? Benign process, often self limited if offending factor removed
  - Non-pharm
    - Modalities (ice), RICE
    - Modify offending activity, pad the elbow, etc
  - Pharm
    - Topical anti-inflammatories
    - NSAIDs (consider side effect profile!)
  - Intervention/surgery
    - Bursectomy? (consider if infected or chronic problem)